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Review

Beyond the Principle of Plentitude: A Review of Terrestrial Planet Habitability

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ABSTRACT

We review recent work that directly or indirectly addresses the habitability of terrestrial (rocky) planets like the Earth. Habitability has been traditionally defined in terms of an orbital semimajor axis within a range known as the habitable zone, but it is also well known that the habitability of Earth is due to many other astrophysical, geological, and geochemical factors. We focus this review on (1) recent refinements to habitable zone calculations; (2) the formation and orbital stability of terrestrial planets; (3) the tempo and mode of geologic activity (*e.g.*, plate tectonics) on terrestrial planets; (4) the delivery of water to terrestrial planets in the habitable zone; and (5) the acquisition and loss of terrestrial planet carbon and nitrogen, elements that constitute important atmospheric gases responsible for habitable conditions on Earth's surface as well as being the building blocks of the biosphere itself. Finally, we consider recent work on evidence for the earliest habitable environments and the appearance of life itself on our planet. Such evidence provides us with an important, if nominal, calibration point for our search for other habitable worlds. Key Words: Extrasolar planets—Habitability—Habitable zone—Planet formation—Terrestrial planets. Astrobiology 5, 100–126.

In some worlds there is no Sun and Moon, in others they are larger than in our world, and in others more numerous. In some parts there are more worlds, in others fewer; in some parts they are arising, in others failing. There are some worlds devoid of living creatures or plants or any moisture.

—Attributed to Democritus (460–370 BCE)

"Nevertheless," she replied, "may these [planetary] systems not, despite their similarity, differ in a thousand ways? After all, a basic resemblance doesn't exclude infinite differences." "Definitely," I answered, "but the difficulty is to figure it out."

—Bernard de Fontanelle, *Entretien sur le Pluralité des Mondes* (1686)

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INTRODUCTION

WHETHER WE ARE ALONE in the universe may be a question as old as human consciousness. Whether there are other planets that support life is a more recent line of inquiry, but by the 17th Century the twin conjectures—that planets exist around other stars and are habitable—were circulating in Western thought (Dick, 1982). In the absence of empirical data or even a theoretical framework, those ideas were based merely on a philosophical "principle of plenitude" (Lovejoy, 1936). The advent of modern geology and biology led to an appreciation of the "physical conditions essential for organic life" on a planet such as the Earth (Wallace, 1903). Multimeter telescopes and robotic spacecraft missions ended speculation that any of the other planets in our Solar System might be Earth-like, but the discovery of extrasolar giant planets and theories of planet formation have provided a scientific basis for the expectation that other stars host Earth-size planets, some of which may be capable of bearing life (Oparin and Fesenkov, 1960). The habitable zone (HZ) was first defined to be the volume around each star in which the stellar flux falls between a proscribed minimum and maximum (Huang, 1959). Successive refinements included calculations of a range of orbital semimajor axes in which a planet's surface temperature is permissive of human habitation (Dole, 1970) or of stable liquid water (Hart, 1979). The main sequence lifetime and luminosity evolution of the parent star and a mass sufficient to retain an atmosphere (but not capture a massive atmosphere) were other criteria for habitability. Kasting *et al.* (1993) used a one-dimensional climate model and included the silicate weathering–carbon dioxide (CO_2) greenhouse feedback that may regulate surface temperatures on the Earth (Walker *et al.,* 1981). They calculated that the HZ around the Sun is conservatively bounded by 0.95 astronomical units (AU), where water is rapidly lost by photolysis and the escape of hydrogen to space, and 1.37 AU, where the formation of CO_2 clouds offsets the greenhouse effect. Rampino and Caldeira (1994) reviewed habitability in terms of climate in 1994, and the proceedings of a conference on the circumstellar HZ were published in 1996 (Doyle, 1996).

Several more recent developments provide further significant insight into planetary habitability. In 1995, the first planet around another main-sequence star was discovered (Mayor and Queloz, 1995), and the number of reported extrasolar (giant) planets has steadily grown to more than 100 (Lissauer, 2002). These gas giant planets illuminate the complexities of planet formation and evolution, and they dynamically constrain the formation and orbits of possible habitable planets in those systems. Exponential growth of desktop computer power and the proliferation of dynamical integration codes have accelerated efforts to model planet formation and orbital evolution. The detection of Earth-sized planets will eventually become feasible with continued improvements in optical engineering and sensor technology (Beichman *et al.,* 2002; Fridlund and Gondoin, 2003). These advances have motivated further studies of planetary habitability, which is increasingly recognized as complex and involving many variables besides orbital semimajor axis (Taylor, 1999). Some of this work appears in a project report (Des Marais *et al.,* 2002) and in the proceeding of a 2000 workshop (Caroff and Des Marais, 2000).

Here, we review the recent (post-1995) literature that addresses either planetary habitability directly or some mechanism thought responsible for the habitability of Earth. We also include some older work that is particularly relevant but not widely considered. We do not consider bodies with subsurface biospheres that cannot be remotely detected (*e.g.,* Europa). Habitable satellites of giant planets, although interesting in their own right (Lorenz *et al.,* 1997; Williams *et al.,* 1997), are not discussed here. We also do not include the detection and characterization of habitable planets in this review as this has been adequately reviewed elsewhere (Woolf and Angel, 1998; Des Marais *et al.,* 2002). Instead, this review is restricted to (1) recent work on the terrestrial planet HZ; (2) the formation and orbital stability of terrestrial planets; (3) geologic activity (*e.g.,* plate tectonics) on terrestrial planets; (4) the origin of terrestrial planet water in the HZ; (5) sources and sinks of terrestrial planet carbon and nitrogen; and (6) evidence for early habitable conditions and life on the Earth.

THE HZ SINCE 1995

In the last decade, HZ calculations have been extended in two ways. Franck *et al.* (2000) calculated the HZ of a "geodynamic" planet with ge-

ologic activity and volcanic CO_2 flux that decreases with time (Franck *et al.,* 1999). They included equations for the thermal evolution of the mantle and seafloor spreading rates (McGovern and Schubert, 1989) and assumed linear growth of continental surface area. A decreased rate of weathering and thus cooler temperatures are required to balance lower volcanic CO_2 at later epochs. Franck *et al.* (1999) included the dependence of weathering rates on biological activity by using the relationships among surface temperature, partial pressure of atmospheric CO_2, rate of plant photosynthesis, soil carbon respiration, CO_2 in soil gas, soil water pH, and weathering rates originally parameterized by Caldeira and Kasting (1992). The HZ of the geodynamic planet spans a greater range of semimajor axes relative to that of the static planet model, but eventually collapses as a result of increasing continental surface area and decreasing volcanic output. However, since silicate weathering is tied to tectonic uplift and exposure of unweathered rock, it should decrease rather than increase with time. Although some of the parameterizations of Franck *et al.* (1999) are *ad hoc,* their work highlights the importance of including biogeochemistry in habitability models.

A second extension is the treatment of non–Earth-like obliquities, eccentricities, and rotation rates. The obliquity of Mars undergoes large and chaotic fluctuations (Touma and Wisdom, 1993) with consequences for climate (Nakamura and Tajika, 2003); without the large angular momentum of the Earth–Moon system, our planet would likely exhibit such behavior (Laskar and Robutel, 1993). Williams and Kasting (1997) used an energy-balance climate model to show that high obliquity would lead to large seasonal temperature variations, particularly in the interiors of high-latitude continents. Nevertheless, storage and transport of heat by the ocean and atmosphere moderate surface temperatures, particularly near the outer edge of the HZ where the planet would have a thicker, CO_2-rich atmosphere. The authors concluded that most Earth-like planets with high obliquity would remain habitable. General circulation model simulations bear out this conclusion (Williams and Pollard, 2003). Williams and Pollard (2002) extended these approaches to Earth-like planets with high eccentricity and came to similar conclusions. The rotation rate of a terrestrial planet may also be an important habitability parameter. Jenkins has examined the effect of rotation on the Earth's early climate in a series of papers, most notably in Jenkins (1996). Faster rotation and larger Coriolis forces inhibit the poleward transport of heat and cool the poles; this could make the climate more susceptible to ice-albedo feedback and runaway glaciation (Budyko, 1969). On the opposite end of the rotation spectrum, Joshi *et al.* (1997) found that a close-in, synchronously rotating terrestrial planet around an M dwarf star could be habitable.

FORMATION AND ORBITAL STABILITY OF TERRESTRIAL PLANETS

The plenitude of habitable planets depends on the frequency of their formation in the HZ and the long-term stability of their orbits. The primary approach to investigating planet formation has been one of numerical simulation, based on a broadly accepted model of terrestrial planet formation, in which gas and dust in a circumstellar disk coagulate for ~0.1 million years (Myr) into kilometer-sized planetesimals. These planetesimals, the building blocks of planets, initiate a comparatively rapid phase of runaway growth that results in lunar-sized embryos, which are gradually perturbed onto crossing orbits by their mutual gravitational interaction (or the presence of a giant planet) and accreted into protoplanets.

Early simulations of terrestrial planet formation and stability were limited in their physical realism. For example, initial work on gas giant planet formation by Pollack *et al.* (1996) did not account for planet migration or dynamical friction. The early models of Wetherill (1996) likewise treated interactions as uncorrelated two-body scattering events. Investigations of terrestrial planet orbital stability have evolved from the simplified models of Gehman *et al.* (1996) to more complex migrating systems such as those simulated by Mandell and Sigurdsson (2003). These multi-body systems are inherently chaotic, and minute variations in initial conditions can lead to drastically different results (*e.g.,* Benest, 1998; Armitage, 2003; Levison and Agnore, 2003). David *et al.* (2003) presented a description of this phenomenon with the initial condition phase angle test depicted in their Fig. 1. One approach to characterizing these chaotic systems is to perform a large number of simulations and generate statistical results such as an expectation value or most probable outcome.

Researchers have also developed increasingly sophisticated numerical techniques to simulate terrestrial planet formation in a more realistic and computationally efficient manner. The studies of Wetherill (1996) and Wetherill and Stewart (1993) used a Monte Carlo technique to simulate perturbations due to close encounters within a swarm of planetary embryos, but did not include long-range gravitational forces between bodies. Noble *et al.* (2002) used a fourth-order Runge–Kutta integration scheme to consider terrestrial stability in the HZ of extrasolar systems. The main limitation of the Runge–Kutta method is that it is confined to short integration times (hundreds or thousands of years). A recent integration tool is the Mean Exponential Growth factor of Nearby Orbits, or MEGNO, indicator (Cincotta and Simo, 2000). This method allows for a computationally efficient distinction between chaotic and regular orbits. Gozdziewski (2002) showed that the MEGNO integration tool yields stability results similar to those of other long-term integrations (Fischer *et al.,* 2002; Laughlin *et al.,* 2002).

The most prevalent simulation tool is the N-body symplectic mapping algorithm originally developed by Wisdom and Holman (1991). Early versions of this tool were incapable of dealing with close encounters between test particles and massive bodies. The SWIFT symplectic integrator developed by Levison and Duncan (1994) solved this problem. SyMBA is another integrator that handles close perihelion passages with short computational time (Duncan *et al.,* 1998). Chambers (1999) developed the MERCURY hybrid sympletic integrator, a tool that also manages close encounters and does not accumulate long-term error. Techniques have evolved such that the planetesimal/embryo transition stage, a previously omitted intermediate step in planet formation, can now be simulated (Barbieri and Marzari, 2002; Quintana *et al.,* 2002). HERMITE, a variation of the symplectic integrator (Makino and Aarseth, 1992), was used to account for gas friction in a planet-forming disk (Kominami and Ida, 2004). This work examined the effect of gas on planetary embryo formation and the importance of the gas lifetime in the solar nebula.

One goal of simulations is to reproduce the number and masses of the terrestrial planets in the Solar System (*e.g.,* Lissauer, 1993; Wetherill and Stewart, 1993; Wetherill, 1996). The majority produce one or more Earth-mass planets near 1 AU. However, Wetherill (1996) found that systems with stars more or less massive than the Sun are 30–50% less likely to contain an Earth-size planet within the HZ. Typically, simulations produce terrestrial planets on orbits with eccentricities of 0.05–0.15, significantly higher than the long-term average of Venus and Mars (Levison and Agnore, 2003); however, those that include a dissipating gas disk do produce planets on low-eccentricity orbits (Kominami and Ida, 2004).

Doppler velocity surveys have revealed that approximately 9% of main sequence stars have giant planets on orbits closer than a few AU (Lineweaver and Grether, 2003). Gravitational perturbations from giant planets can affect the formation of terrestrial planets within the HZ by accelerating planetesimals to crossing speeds where destruction rather than aggregation occurs, inhibiting embryo formation, but also by increasing the eccentricity and orbit crossings of embryos, promoting formation of protoplanets (Whitmire *et al.,* 1998). Thèbault *et al.* (2002) showed that rapid formation of the giant planets detected in the 47 Ursa Majoris and Epsilon Eridani systems could preclude the accretion of planetesimals into embryos. This would be an issue if these giant planets rapidly formed via a disk instability mechanism (Boss, 1997) but not if giant planets formed around a slowly accreting solid core (Mizuno, 1980). Weidenschilling (2000) demonstrated that perturbations by gas giants increased the eccentricity of embryos, creating larger (Earth-sized) planets in the HZ. Simulations performed by Levison and Agnore (2003) showed that terrestrial planet formation is likely if gas giants are no closer than ~4 AU to the central star. The presence of gas giants leads to fewer and more massive planets, and, conversely, systems without a gas giant usually result in a larger number of smaller planets (Fig. 1).

Orbital migration is invoked to explain the significant fraction of gas giants found close to their central stars, *e.g.,* by Trilling *et al.* (2002). Kortenkamp and Wetherill (2000) showed that perturbations of Jupiter and Saturn would have been considerably smaller if they had drifted inward from more distant original positions. Kominami and Ida (2004) reported that without inward radial migration, the present positions (and eccentricities) of these giant planets would have precluded damping of terrestrial planet eccentricities by a dissipating gas disk. On the other hand, giant planet migration through the zone of terrestrial planet formation is destructive; Armitage

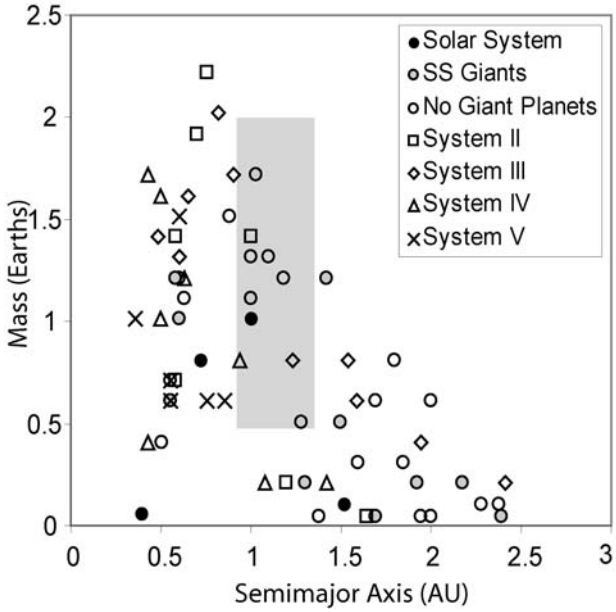

FIG. 1. Terrestrial planet semimajor axes (AU) and masses (Earths) from the simulations of Levison and Agnore (2003). Four independent simulations were carried out for each configuration of giant planets, and the terrestrial planets produced in each simulation are shown. The shaded area represents the HZ of the Sun as defined by Kasting *et al.* (1993) and assuming that a habitable planet must have a mass within a factor of 2 of the Earth. Solid black points represent the actual Solar System. Gray filled points represent the result of four simulations with the outer Solar System planets in the current configuration. Open circles represent a system with no giant planets. Other symbols represent systems with different configurations of giant planets.

(2003) found that the material swept up by a giant planet would not be replenished if migration happened after the earliest stage (~0.1 Myr) of planet formation.

About two-thirds of stars are members of multiple systems. Companion stars would influence planet formation in the same manner as giant planets, but at yet larger distances. Whitmire *et al.* (1998) studied the conditions under which perturbations by a companion star would prevent runaway accretion of planetesimals in the HZ. They derived criteria based on binary semimajor axis (a), eccentricity (e), and stellar masses. For example, habitable planets will form in a 2 solar mass system with an eccentricity of 0.5 only if $a >$ 32 AU. Using the empirical distribution of binary orbital elements, they concluded that 60% of solar-type binaries could host habitable planets. More recent studies of Alpha Centauri (a = 23.7 AU, e = 0.519) would appear to contradict the empirical relation of Whitmire *et al.* (1998) (which predicts that planets would not form in the HZ). Barbieri and Marzari (2002) found that planets would be precluded only outside 3 AU. Ironically, this may preclude giant planet formation and remove a potential source of destructive perturbations. Quintana *et al.* (2002) showed that two to five terrestrial planets can form from a disk inclined as much as 45° with respect to the plane of a binary system.

There has been little examination of other factors that might influence terrestrial planet formation. Agnor *et al.* (1999) showed that the rotation of the Earth was a result of multiple giant impacts. The stochastic nature of these events means that terrestrial planets may exhibit marked kinematic diversity. Lineweaver (2001), among others, has proposed that the abundance of heavy elements in a planet-forming nebula is important to the final outcome, and he estimated that the range of abundance over which the probability of a star harbors an Earth is finite, 0.1 to three times the solar value, bounded at the lower end by the ability to form a planet and at the higher end by the migration of giant planets into the HZ.

Studies of the orbital stability of terrestrial planets are a natural extension to research on their formation. The primary impetus is to determine whether terrestrial planets in the HZ are dynamically stable on time scales comparable to the emergence of life (see above). Simulations of terrestrial planet orbital stability have focused on (a) systems with known giant planets or (b) hypothetical planetary systems. One criterion for orbital stability is the Hill radius, the distance within which a neighboring planet's gravity dominates over the parent star. Test particles are considered unstable if they experience an encounter with another body closer than the Hill radius. Some groups adopt stability criteria that are not analytically derived but can be easily incorporated into numerical simulations. For instance, Holman and Wiegert (1999) checked for close encounters between the components of a stellar binary system and test particles by defining a close encounter as the distance equal to one-quarter of the periastron separation of the binary. David *et al.* (2003) defined stability criteria with numerical limits to the energies, eccentricities, and semimajor axes of their test particles.

One of the earliest works on the orbital stability of potentially habitable planets in known extrasolar planetary systems is that of Gehman *et*

al. (1996). One of the systems they examined is 47 Ursa Majoris. This is a multiple giant planet system like our own Solar System (Fischer *et al.*, 2002). Rather than employing numerical simulations, Gehman *et al.* (1996) used an analytical method and assume a massless terrestrial planet on a nearly circular orbit co-planar to that of the known giant planets. They concluded that the system is likely to be dynamically habitable for an Earth-like planet. Laughlin *et al.* (2002) ran formation models and found that it would be unlikely for Earth-like planets to form within this system. However, they also concluded that if a terrestrial planet did form, it would be dynamically stable. Gozdziewski (2002), using the MEGNO integration tool, found that this system has several narrow zones of stability within the HZ that are confined by mean motion resonances. Dynamical analogy was drawn between the HZ of 47 UMa and the asteroid belt of the Solar System. Jones and Sleep (2002) reached the same conclusion using a symplectic integrator, and found that only 10% of the HZ is stable. Cuntz (2002) employed an integrated system approach to constrain the HZ, incorporating both a geodynamical model of CO_2 abundance and a planetary climate model. He found that it is unlikely for a terrestrial planet to be dynamically habitable. Nevertheless, he concluded that of the discovered extrasolar systems, 47 UMa is the most promising candidate to harbor a habitable terrestrial planet. Menou and Tabachnik (2003) simulated the dynamical habitability of all known extrasolar planetary systems (85 at the time the work was performed). These simulations involve placing 100 test particles into the HZ of a given system and letting their orbital elements evolve for 1 Myr. The number of remaining particles is an indicator of relative orbital stability within the HZ. Their numerical results confirmed intuition; systems with the least amount of overlap between the HZ and the gas giant region of gravitational influence are the most dynamically habitable (Fig. 2).

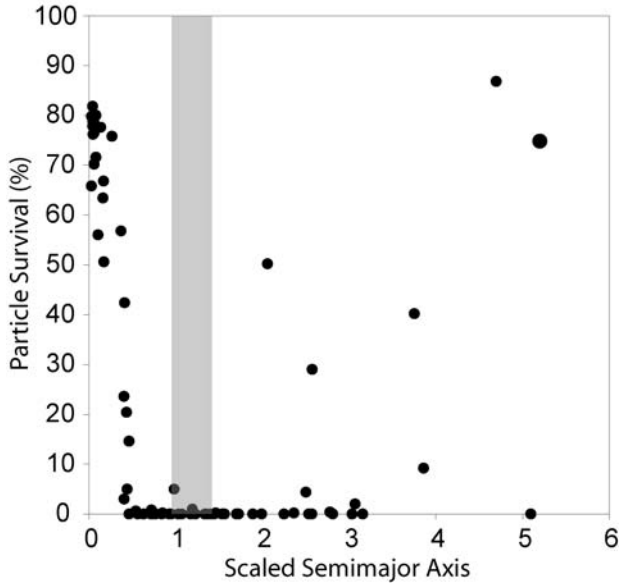

FIG. 2. Plot summarizing the dynamical habitability of known extrasolar systems as determined by the simulations of Menou and Tabachnik (2003). Each point represents the outcome of one simulation in which test particles are distributed uniformly throughout the HZ and the percent survival recorded after 1 Myr of simulation time. The semimajor axis of the (known) giant planet is scaled by the HZ (relative to that of the modern Sun). The large solid point is the Solar System, and the shaded region is the HZ as defined by Kasting *et al.* (1993).

Another approach to investigating dynamical habitability is to explore some of the vast parameter space that a hypothetical system could occupy. In a series of papers spanning 15 years, Benest and co-workers have investigated the stability of test particles in a number of binary star systems (Benest, 1988, 1989, 1996, 1998, 2003). Benest considered only a single planet on a coplanar, s-type orbit (about one star). From these relatively simple experiments he concluded that stable planetary orbits exist at radii of roughly half the binary's periastron separation. In a pair of publications in 2003, Dvorak *et al.* (2003a,b) explored the dynamics of the binary star system Gamma Cephei, which has recently been discovered to harbor an s-type gas giant, and the HD 74156 system, in which two gas giants are known to reside, one of which orbits within only 0.28 AU of the parent star. By adding a terrestrial planet to Gamma Cephei they found that even this fairly complex system has regions of stability within the HZ of the system. The second work focused on resonances between a hypothetical terrestrial planet and the two giants. They found that nearly all test particles initially placed at distances corresponding to resonant orbits are unstable. Furthermore, inward migration of the inner gas giant (and its resonances) to its current orbital radius would make HD 74156 an even more unfavorable environment for a terrestrial planet.

Dvorak and Süli (2002) ran a series of simulations in which they varied the mass of the terrestrial planets (Venus, Earth, and Mars) in a sim-

ulated solar system by a common multiplicative factor. They found that even when the masses are increased by a factor of 220 the system remains stable over tens of millions of years, the duration of their simulations. David *et al*. (2003) employed a novel approach to determine the fraction of binary systems that are dynamically habitable over 4.7 billion years, the approximate age of the Solar System. They derived an empirical relation between the ejection time for a terrestrial planet at 1 AU and the periapsis of the binary system (either a star and a gas giant or two stellar mass objects) from numerical simulations. Combining this with the known periastron distribution of stellar binaries, they estimated that 50% of Earth-like planets in binary systems will be dynamically stable over 4.7 billion years. However, many of the simulated planets do not remain in the HZ, and thus the actual fraction of continuously habitable systems will be lower. Finally, Lissauer (2001) performed a unique series of simulations in which an extra planet is added to the solar system in place of the asteroid belt. He found that this body increases the exchange of angular momentum between the gas giant and the inner planets, thus increasing the chance of ejection for each of the terrestrial planets. However, even with the enhanced probability of ejection, this study shows that planets can remain stable over time scales of at least 1 billion years, *i.e.,* long enough for life to emerge.

The effect of gas giant migration on terrestrial planet stability has only recently been considered, though it is likely that terrestrial planet survival probability depends on migration rate. Mandell and Sigurdsson (2003) simulated the effect of the migration of a Jupiter-like body through a hypothetical inner solar system. They found that in 1–4% of their simulations terrestrial planets can remain within the HZ as the gas giant migrates through this region. They also suggested that the orbit of a terrestrial planet perturbed by a migrating gas giant could be circularized by interaction with a remnant gas disk.

As discussed previously, the final rotation and obliquity states of terrestrial planets are important factors for habitability. The rotation state of the Earth has been investigated in great depth and has been reviewed by Lissauer *et al*. (2000). The high angular momentum of the Earth–Moon system, possibly a result of an oblique Moon-forming impact, separates spin precession frequencies from orbital precession frequencies and stabilizes the Earth's obliquity (Laskar and Robutel, 1993). Little work has been done to simulate the distribution of rotation rates and obliquities of terrestrial planets and their subsequent evolution under the influence of possible spin-orbit resonances. Atobe *et al*. (2004) considered the evolution of terrestrial planet obliquities and the effects of spin-orbit resonances in known extrasolar planetary systems. They found that ~20% of known extrasolar planetary systems can harbor terrestrial planets that have dynamically stable orbits and experience obliquity fluctuations less than 10°, conditions considered favorable for habitability.

TERRESTRIAL PLANET GEOLOGY

Geologic activity underpins many of the processes responsible for the habitability of Earth. The long-term carbon cycle (discussed below) includes volcanic and metamorphic degassing, weathering of uplifted silicate rocks, and burial of organic carbon and carbonates in sedimentary basins and passive continental margins. In the absence of geologic activity and balance by volcanic outgassing, weathering of silicate rocks would remove all carbon from the modern atmosphere-ocean and sequester it as carbonates in 1 Myr (Schlesinger, 1997). The amount of calcium required is released in the complete weathering of a rock layer only 10 m thick. Thus silicate weathering would not be limited in the absence of tectonic orogeny and would halt only at globally freezing temperatures (Walker *et al.,* 1981). The magnetic dipole field produced by Earth's convecting liquid metal core and that protects the atmosphere from sputtering by the solar wind, particularly during the early, more active history of the Sun (Yung and DeMore, 1999), is another manifestation of geologic activity important to habitability. New theoretical insights into the mechanisms and modes of geologic activity on terrestrial planets have benefited from geophysical data returned by robotic probes to Venus (Magellan) and Mars (Mars Global Surveyor).

In the modern geological paradigm, activity on the Earth's surface is intimately linked to convection in its interior. All terrestrial planets and even the Moon have interiors that convect as they lose heat from the decay of radioisotopes (especially the long-lived isotopes ^{40}K, ^{232}Th, ^{235}U, and ^{238}U), mantle cooling, and any latent heat from a solidifying core. The abundance of radiogenic iso-

topes and thus the amount of heat produced in planet-forming material may vary with time and space in the Galaxy. These unstable isotopes are produced by the *r*- and *s*-processes in massive stars and released into the interstellar medium by type II supernovae. Gonzalez *et al.* (2001) calculated the abundance of these isotopes relative to Fe, a major planet-building element, and predicted the relative radiogenic heat production in the interiors of planets formed at different epochs. They found a secular decrease in radiogenic heating with time, such that a planet formed at the present epoch will have only 60% of Earth's radiogenic heating. This conclusion may be in part an artifact caused by scaling with Fe, an element produced primarily in type I supernovae and whose relative abundance with respect to other planet-forming elements has increased over galactic history (Timmes *et al.*, 1996). Long-lived radioisotope abundances also vary significantly between the different classes of chondritic meteorites, perhaps reflecting condensation in the protosolar nebula (Newsom, 1995).

Convection occurs in a fluid when the Rayleigh number, the dimensionless ratio of buoyant to viscous forces, exceeds a critical value. The Raleigh number of an internally heated fluid of depth h scales as Qh^5, where Q is the specific heat production (Turcotte and Schubert, 1982). In the case of a cooling body such as a planet, h is the thickness of the mantle, and Q would include the heat released during cooling. The thickness at which mantle convection occurs (a few hundred kilometers) is thus relatively insensitive to Q, and convection will be a universal property of terrestrial planets regardless of their exact inventory of radiogenic isotopes. Likewise, the thermal history of a planet will be much more sensitive to the size of the planetary body (Spohn, 1991), which is consistent with the observed relative rates of geologic activity on surfaces in the inner Solar System.

Although mantle convection may be universal, recent theoretical work suggests that the mode of mantle convection and the involvement of the lithosphere and surface can vary dramatically between planets. Scaling arguments and numerical models delineate three distinct modes: a plate tectonic regime (Earth), a stagnant lid regime where convection occurs beneath a stable brittle lithosphere (Moon, Mercury, and Mars), and an intermediate regime in which the lithosphere episodically founders (Venus) (Moresi and Solomatov, 1998). The mode depends on Raleigh number and the viscosity contrast across the mantle. The exponential temperature dependence of mantle material produces a large viscosity contrast between the surface and interior, which inhibits plate tectonics in favor of a stagnant lid. The occurrence of plate tectonics on the Earth and its absence on Venus (where the temperature contrast is smaller), however, demands another explanation. One hypothesis is that shear stresses in the upper mantle exceed the yield strength of Earth's lithosphere, set by the strength of faults. Moresi and Solomatov (1998) found that the requirement of weak faults corresponds to a friction coefficient $\mu < \omega\alpha\Delta T$, where ω is between

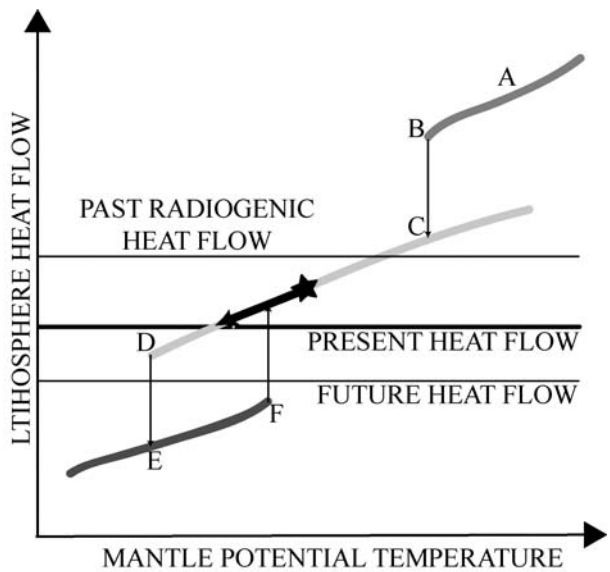

FIG. 3. Conceptualization by Sleep (2000) of the thermal and geologic evolution of a terrestrial planet as a trajectory in the space described by mantle potential temperature and lithosphere heat flow. The red, green, and blue curves describe a magma ocean, plate tectonics, and a stagnant lid mode of mantle convection, respectively. Earth is at the position marked by a star, and its trajectory is along the plate tectonic curve to the left because the lithosphere heat flow exceeds the modern radiogenic heat flow and the mantle is cooling. A terrestrial planet with an impact-generated magma ocean is at point A. Because the planet's past radiogenic heat flow was insufficient to maintain a magma ocean, the planet evolves along the magma ocean curve to point B, where it makes a catastrophic mode switch to plate tectonics at point C. Plate tectonics will continue to operate until the planet has cooled to point D, whereupon it switches to a stagnant lid mode at point E. If the future radiogenic heat flow exceeds the flow that can be conducted through a stagnant lid, the planet reverts to plate tectonics at point F. The evolution of any given planet will depend on the radiogenic heat flow (and thus planetary size) and the range of mantle temperature and lithosphere heat flow over which the plate tectonic mode extends.

1.25 and 1.75, α is the coefficient of thermal expansion of mantle material, and ΔT is the temperature drop across the mantle. The critical value of μ is between 0.03 and 0.13. Foster and Nimmo (1996) measured larger graben widths on Venus compared with Earth and deduced that the strength of faults on the former is 1–2 orders of magnitude stronger. Water may weaken faults on the Earth (but not on Venus), and allow plate tectonics to occur (Nimmo and McKenzie, 1998).

Alternatively, plate tectonics may be limited by the magnitude of shear stresses in the mantle. Lenardic *et al.* (2004) calculated the effect of a thick, buoyant, non-subducting crust on a planet with active plate tectonics. The decrease in area of oceanic crust through which heat can be efficiently conducted leads to an increase in the temperature of the mantle and a decrease in viscosity until the shear stresses no longer exceed the yield strength of the lithosphere. They explained the hemispheric dichotomy in martian geology as a consequence of a termination of early plate tectonic activity (Sleep, 1994) when the southern cratered highlands grew to cover half the planet. However, their model also predicts that plate tectonics would have re-initiated on Mars as the mantle cooled, and that Earth would have operated in the stagnant lid regime in the past, both in conflict with observations.

Sleep (2000) presented a conceptual model for modes of mantle convection that seeks to explain the different evolutionary histories of the terrestrial planets. He made the "large planet" assumption that a planet's thermal state can be uniquely described by the potential temperature of the convecting mantle and the heat flow through the lithosphere, and he described evolution in terms of trajectories in temperature–heat flow space (Fig. 3). Three modes of mantle heat transport (magma ocean, plate tectonics, and stagnant lid) are described by individual curves, and catastrophic jumps between modes can occur. He proposed that the temperature range of plate tectonics is limited by the disappearance of mantle melting and cessation of seafloor spreading at low temperatures, and greater mantle melting and formation of crust too thick to subduct at high temperatures. The Earth's mantle may have cycled between a magma ocean and plate tectonics in the past, and may cycle between the stagnant lid mode and plate tectonics in the future. On Venus, the plate tectonic mode is inaccessible (perhaps because of lack of water), and cycling occurs between magma ocean and stagnant lid regimes. Because of its smaller size and lower heat flow, Mars may have only transiently passed through the plate tectonic regime early in its history (Sleep, 1994). Likewise, Earth-sized planets with a smaller inventory of radiogenic isotopes would also exhibit plate tectonic behavior in their early history.

Could a planet maintain significant geologic activity in the absence of plate tectonics? One possibility is "hot spot"-style volcanism, widely accepted to be a consequence of buoyant plumes ascending from deep within the mantle, perhaps from the core–mantle boundary. Hot spot volcanism accounts for only a small fraction of degassing from the modern Earth; however, it may have been important on Venus and early Mars, and contributed to surface volatile budgets (Phillips *et al.,* 2000). Release of heat from a cooling, crystallizing core could drive hot spot volcanism. However, cooling of the core may in turn depend on effective cooling of the mantle by plate tectonics.

The geology of rocky planets will also depend on the degree of melting that occurs at sites of mantle upwelling (*e.g.,* beneath mid-ocean ridges and at plumes). Melting occurs where the adiabat of ascending mantle material crosses the solidus. In the absence of significant melting, ultramafic mantle material would be directly emplaced on the seafloor. The best terrestrial analogs to this regime are probably the ultraslow-spreading sections of ocean ridges, where the lower Peclet number (ratio of advective to conductive heat transport) is predicted to result in decreased melt production, intermittent volcanism, thin crust, and emplacement of mantle to the seafloor (Dick *et al.,* 2003). Recent surveys of Gakkel Ridge in the Arctic Ocean, which is the slowest spreading portion of the global mid-ocean ridge system, have found evidence for decreased, heavily focused melt production and a relatively thin crust (Coakley and Cochran, 1998; Edwards *et al.,* 2001; Jokat *et al.,* 2003; Michael *et al.,* 2003). Two plausible consequences of decreased mantle melting would be attenuated degassing to the atmosphere-ocean and a lower amount of eclogite (produced by metamorphism of basalt) in subducted oceanic plates. High-density eclogite may contribute to the "slab pull" mechanism driving plate motion. The solidus is affected by the Mg/Fe ratio, which may vary between planetary systems. Water exerts a more profound effect on

melting of silicates; the solidus of a "wet" mantle occurs at a significantly lower temperature than it does for a "dry" mantle. Water increases the total melt production while decreasing the mean extent of melting (melt fraction) (Asimow and Langmuir, 2003).

WHENCE THE WATER?

The necessity of water for planetary habitability is widely accepted. Earth's water has been an intimate component of biochemistry, the primary constituent of cellular cytoplasm, a habitat for early life, and a sink for atmospheric CO_2 that regulates the greenhouse effect of that gas. As discussed previously, water plays an important role in geologic activity. Identifying the origin of Earth's water is necessary for understanding the factors that control the abundance of water on terrestrial planets in other HZs.

Water was the most abundant condensate in the outer solar nebula where it could have triggered giant planet formation (Stevenson, 1988) (whose consequences have been discussed). However, models of the solar nebula predict high mid-plane temperatures, leaving planet-forming material at Earth's orbit—and in the entire HZ—relatively depleted of volatiles such as water (Stevenson, 1988; Cyr *et al.,* 1999). These predictions are supported by the low water content (0.1% by weight) of enstatite chondrites, meteorites derived from parent bodies thought to have formed within the inner nebula. Earth's ocean may have arisen from impact degassing of planetesimals during accretion. Matsui and Abe (1986) showed that the mass of the hydrosphere is controlled by the solubility of water in silicate melts and that even source material with a low water content (0.03%) is capable of supplying a mass of water equivalent to the modern ocean (0.023% of an Earth mass). However, a large amount of water may have been lost as hydrogen from the early Earth (Pepin, 1991). Thus, researchers have proposed additional sources of water, particularly from the water-rich outer Solar System.

One possible source is a post-accretion veneer of material, *e.g.,* asteroids from the main belt (Boato, 1954). A second proposal is that the main contributor was comets (Chyba, 1987). A third is that ice particles in the outer regions of the primordial Solar System experienced drag by nebular gas and drifted inward to the terrestrial planet zone (Cyr *et al.,* 1998). A recent theory holds that planetary embryos from beyond 2.5 AU were the source of Earth's water (Morbidelli *et al.,* 2000). Any successful theory must explain the terrestrial abundance of water [9×10^{22} mol or 2.8×10^{-4} Earth masses at the surface and in the crust and $0.8–8 \times 10^{-4}$ Earth masses in the mantle (Lécuyer, 1998)] and the ratio of deuterium to hydrogen isotopes, or D/H, in seawater (1.53×10^{-4}). The isotopic composition of the ocean has evolved over geologic time as water exchanges between the mantle and ocean at zones of seafloor spreading and subduction, extraterrestrial water is accreted, and H atoms escape to space. Lécuyer *et al*. (2000) modeled that evolution and estimated that the H isotopic composition would have changed by only −20‰ to +10‰ over the past 3.5 Ga, which is insignificant compared with chemical variation in the Solar System. They concluded that the D/H of modern seawater is still informative about the source of Earth's water.

Carbonaceous chondrite (CC) meteorites contain abundant water (1–10% by weight) with a D/H ratio similar to the terrestrial value. A late veneer of CC material could explain mantle concentrations of highly siderophilic elements (Ni, Co, Ge, and the platinum group elements) that strongly partition into Fe melt and should have been removed from the mantle during core formation (Kimura *et al.,* 1974). Although the anomalous siderophile abundances might also be explained by higher partition coefficients at the high pressures found at the base of a primordial magma ocean (Righter and Drake, 1997), this interpretation has been recently challenged with new data (Holzheid *et al.,* 2000). Resolution of this issue awaits further experimental and analytical work (Righter, 2003).

Comets have been proposed as another source of Earth's water (Chyba, 1987). Since 1995, D/H has been measured in three long-period comets: P/Halley (Balsiger *et al.,* 1995; Eberhardt *et al.,* 1995), C/1996 B2 (Hyakutake) (Bockelée-Morvan *et al.,* 1998), and C/1995 O1 (Hale-Bopp) (Meier *et al.,* 1998). The isotope ratios are twice that of standard mean ocean water (SMOW), leading most investigators to conclude that comets were a minor contributor to the terrestrial water inventory. However, Delsemme (1999) argued that the source of Earth's water was a now-extinct population of icy planetesimals from the Jupiter zone, rather than long-period comets from the

Oort cloud. He proposed that D/H enrichment in the latter arises in the icy mantles of interstellar grains that experienced ion-molecular chemistry at low temperatures. Comets from the Oort Cloud are thought to originate primarily from the Uranus–Neptune region where low temperatures would have quenched deuterium exchange. In contrast, any D/H enrichment in water in the warmer Jupiter region of the nebula would have been erased as water exchanged deuterium with nebular hydrogen having a primordial D/H ratio one-sixth the terrestrial value. Dauphas *et al.* (2000) assumed that terrestrial water originated from a mixture of Earth-forming planetesimals, asteroids, and comets. They constrained mass-balance equations with D/H values, the lunar impact record, and terrestrial mantle siderophile abundances to conclude that late-accreting asteroids were the dominant (50%) source of water and comets contributed less than 10%. The remaining water would have been included in the proto-Earth.

Cyr *et al.* (1998) proposed that the inner solar nebula was replenished with water by inward drifting ice particles. The Keplerian velocity of solid bodies would have exceeded that of the pressure-supported nebular gas, and the resulting gas drag would have caused particles 0.01–10 m in size to drift inward. Icy bodies drifting inwards of 5 AU would have sublimated in the warmer inner nebula. Outward diffusion then returned the water (as vapor) to the outer solar nebula where it re-condensed. This movement of water would have increased the steady-state concentration available for incorporation into planetesimals and the proto-Earth between 0.1 and 2 AU. Kuchner (2003) proposed an even more radical mechanism to return water to the HZ: He suggested that terrestrial planets may start as icy bodies in the outer regions of a planet-forming disk and migrate inward.

An even more recent hypothesis is that Earth received its inventory of water by accreting one or more planetary embryos from more distant orbits. Morbidelli *et al.* (2000) proposed that "wet" planetary embryos formed beyond 2 AU, and that after Jupiter formed, its gravitational perturbations increased the eccentricity of these embryos, leading some to collide with the terrestrial planets. They carried out simulations in which these are accreted in the last stage of planet formation, allowing for greater retention of water. In contrast with the Earth, Mars may be small enough to constitute a single planetary embryo, and the bulk of its water could have been from smaller asteroids and comets (Lunine *et al.*, 2003). Their simulations show that Mars could have received an amount of water equal to 6–27% of Earth's present ocean with a D/H 1.2–1.6 times the terrestrial value, which is consistent with estimates of the inventory of water based on geomorphology and the D/H ratio in SNC (martian) meteorites. Raymond *et al.* (2004) carried out a series of simulations in which they model the formation of Earth-sized planets in the HZ and the incorporation of water into those planets. The mass, semimajor axis, and orbital eccentricity of a giant outer planet, the position of the water condensation front in the nebula, and the density of solids in the nebula are varied between simulations. The authors assumed that only those bodies outward of 2.5 AU contain water. Of the 45 planets that formed between 0.8 and 1.5 AU, 35 received an inventory of water equal to or greater than that of the Earth (Fig. 4).

Of course, the long-term habitability of a terrestrial planet will depend not only on the initial inventory of water but also the long-term reten-

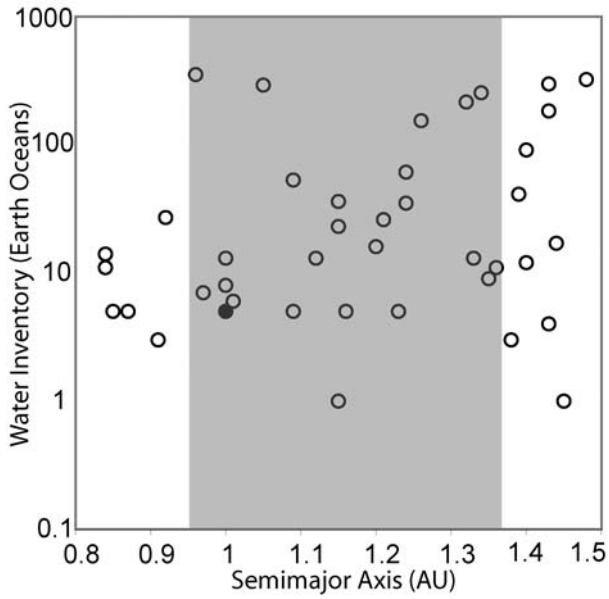

FIG. 4. Final water inventory (in units of Earth oceans) versus semimajor axis of Earth-size planets produced by the simulations of Raymond *et al.* (2004). The Earth (with an estimated initial inventory of five oceans) is the solid point. Four planets with negligible estimated water inventory are not shown in this logarithmic plot. The shaded region is the HZ as defined by Kasting *et al.* (1993).

tion of that water against loss to space or to the interior of the planet. Water and iron at high pressures form iron hydride (FeH_x), and the reaction of metallic Fe with the hydrous mantle during core formation could sequester a significant amount of H in the core. Hydrogen is one candidate for the light element responsible for the discrepancy between the laboratory-determined density of pure iron and the seismically determined density of the core. Hirao *et al.* (2004) reported new data on the equation of state of FeH_x to 80 GPa and estimated that the core could contain 0.12–0.48% by weight H, equivalent to eight to 24 oceans of H_2O, less than previous estimates, but still large (the implications of an alternative light core element—carbon—are discussed in the next section).

Volatiles such as water also strongly partition into silicate melts; thus the total rate of mantle melting controls the rate at which juvenile water is released from the mantle. Water is returned to the mantle by incomplete dewatering during metamorphism of subducted oceanic slabs. In a series of papers based on earlier work (McGovern and Schubert, 1989), Franck and colleagues estimated the fluxes of water from a mid-ocean ridge source and to the subduction zone sink, and included both modern and future water loss to space (Franck and Bounama, 1995, 1997; Bounama, 2001). They predicted that 1 billion years from now only one-quarter of Earth's water will be returned to the mantle and concluded that feedback mechanisms between sinks and sources will prevent complete absorption of surface water into the mantle. The ultimate loss of Earth's water will occur by increasing solar luminosity, the onset of a wet greenhouse in 1.5 billion years, photodissociation of atmospheric water vapor, and total loss of H to space 1 billion years later (Caldeira and Kasting, 1992).

OTHER ELEMENTS OF HABITABILITY

Carbon and nitrogen are also critical to life and terrestrial planet habitability. Livio (1999) estimated the cosmological production of carbon as a constraint on the emergence of carbon-based life in the universe. However, C and N are not only the building blocks of life on Earth, but also form incondensable atmospheric gases that maintain surface pressures and temperatures above the triple point of water (6.1 mbar and 273.25 K) and shield the surface against high-energy radiation from space. The key atmospheric gases on modern Earth are dinitrogen (N_2) and CO_2. N_2 is an inert buffer gas that permits gas exchange between life forms, and pressure-broadens the infrared absorption lines of greenhouse gases, amplifying their effect. CO_2 is the major greenhouse gas on the modern Earth; methane (CH_4) may have been as important in the Archean (Pavlov *et al.*, 2000). The exchange of these atmospheric gases with other surface reservoirs (hydrosphere, sediments, and crustal rocks) affects long-term planetary habitability. The temperature-dependent removal of inorganic carbon from the ocean-atmosphere as carbonates may have stabilized Earth's CO_2 greenhouse climate against the luminosity evolution of the Sun (Walker *et al.*, 1981). The C and N involved in these geochemical cycles exceed that in living matter by a factor of 10^6 (Fig. 5).

The incorporation of C and N in primordial Solar System solids and the Earth was very rela-

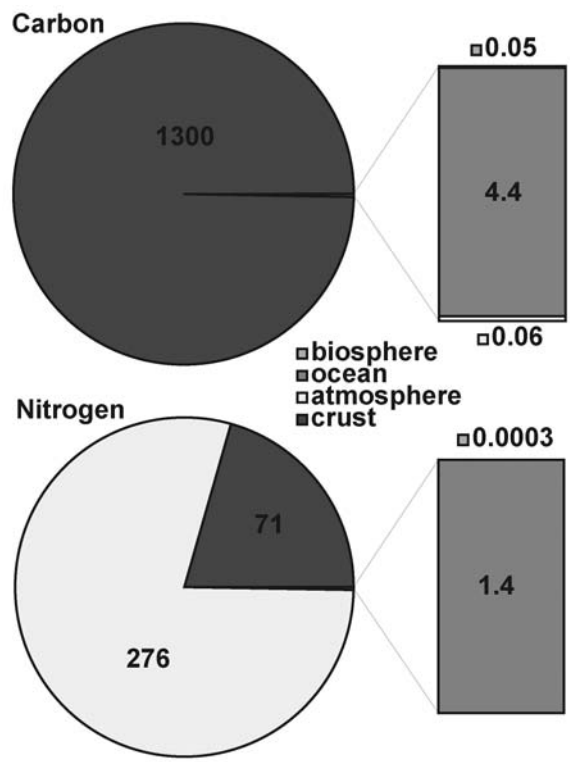

FIG. 5. Surface inventories of carbon (top panel) and nitrogen (bottom panel) on the Earth in units of 10^{18} mol. The oceanic reservoir includes both inorganic and organic carbon and dissolved N_2. Data are from Schlesinger (1997).

tively inefficient. For example, when compared with silicon (Si), C and N are depleted in primitive chondrites by factors of 25 and 130, respectively, relative to solar abundances. They are depleted yet again by factors of 60 and 400 relative to elemental abundance estimates of the bulk modern Earth (Newsom, 1995). The other terrestrial planets in our Solar System probably started with similar complements of C and N, but we cannot be certain of this. The martian meteorites are basaltic and cannot be considered representative of the bulk planet. Lécuyer *et al.* (2000) calculated that the Venus atmosphere contains twice as much carbon and nitrogen as the telluric surface inventories; however, they ascribed this difference to the storage of C and N in the mantle by terrestrial plate tectonics.

The prevailing view is that primitive meteorite-like material was the source of planetary C and N in the Solar System. The isotopic composition of C in the terrestrial mantle ($\delta^{13}C = -5‰$) and in martian meteorites ($-7‰$) falls within the range of whole rock measurements of CC meteorites [-25 to $0‰$, values that result from an admixture of isotopically light organics and a minor contribution of isotopically heavy carbonate and labile organic phases (*e.g.,* Sephton *et al.,* 2003)]. Likewise, the isotopic composition of N in the acid-insoluble organics of CC meteorites is similar to that of the Earth (Alexander *et al.,* 1998). However, a cometary contribution to terrestrial N has been suggested as corollary to the theory that fractionation of rare gases in cold cometary ices can explain the high terrestrial ratio of krypton (Kr) to xenon (Xe) relative to meteorites (Owen *et al.,* 2001). Dauphas (2003) successfully described the terrestrial rare gas elemental and isotopic abundances by mixing meteorite and cometary volatile end-members, but he estimated that comets contribute $<3\%$ of the terrestrial N inventory. A few N isotope measurements have been made in CN and HCN from comets. Jewitt *et al.* (1997) reported a $\delta^{15}N$ ($\delta^{15}N = [(^{15}N/^{14}N)/(^{15}N/^{14}N)_{Earth} - 1] \times 1{,}000$) of $-160 \pm 100‰$ in hydrogen cyanide (HCN) of Comet 1995 O1 (Hale-Bopp). Arpigny *et al.* (2003) found $\delta^{15}N = 900 \pm 500‰$ in the cyanide radical (CN) in both Hale-Bopp and Comet 2000 WM1 (LINEAR). Cometary CN derives partly from HCN; thus the two Hale-Bopp measurements imply the existence of an isotopically heavier CN parent such as the organic molecules found in some primitive meteorite grains. ^{15}N-enriched cometary organics would seem an unlikely source of terrestrial planet N.

Maurette *et al.* (2000) identified a third possible source of terrestrial planet C and N based on their analyses of micrometeorites or interplanetary dust particles recovered from Antarctic ice. Their interplanetary dust particles are elementally similar to CM-type (Murchison) meteorites but are distinct in terms of pyroxene to olivine ratio, rarity of chondrules, and ratio of α-isobutyric to isovaline amino acids. They proposed that they are primitive solar system material not currently represented in meteorite collections. Their micrometeorites contained an average 2.5 wt% C (similar to CM chondrites) but an N/C ratio (≤ 0.04) that resembles comets. They compared the volatile contribution of interplanetary dust particles (estimated by scaling the modern flux to the lunar impact history) with current terrestrial inventories of water, C, and N. The agreement for C and N is almost certainly coincidental, given the many assumptions required. A final source of terrestrial C and N may have been dissolution of a gravitationally captured, solar-composition atmosphere into a magma ocean created by giant impacts or core formation, and later degassing from the mantle. Such a process has been invoked to explain isotopically solar-like neon in the Earth's mantle (Porcelli and Pepin, 2000). However, new measurements of solar isotopic abundances cast doubt on a solar origin of Earth's C and N (see below).

The origin of the organic phases in primitive meteorites is reviewed by Ehrenfreund and Charnley (2000), and only recent work is highlighted here. Carbon monoxide (CO) is the dominant carbon species in the interstellar medium and was thermodynamically stable throughout most of the protosolar nebula. Possible sources of organic molecules are (1) interstellar organics, (2) Fischer–Tropsch conversion of CO and H_2 into CH_4 and heavier hydrocarbons on the surfaces of metal grains, and (3) photochemical processing of nebular gas. Deuterium enrichment in meteorite organics is indicative of a low-temperature, interstellar origin and rules out a significant contribution by aqueous chemistry in the parent bodies of meteorites where the D/H values of H_2O are close to terrestrial [Sandford *et al.* (2001) discussed mechanisms of D-enrichment in polycyclic aromatic hydrocarbons, an important organic component in CC meteorites]. However, interstellar organics are subject to evaporation

within the warm inner nebula; recent evaporation experiments using interstellar organic analogs suggest that these cannot account for the C abundance of CC meteorites (Nakano *et al.,* 2003).

Fischer–Tropsch synthesis corresponds to reactions of the form

$$(n + 0.5m)H_2(g) + nCO(g) \rightarrow C_nH_m(s) + nH_2O(g)$$

and has long been considered a plausible source of organics under solar nebular conditions (Prinn and Fegley, 1989), but this mechanism cannot explain the enrichments of D and ^{15}N in meteorite organic matter. The formation of polycyclic aromatic hydrocarbons and normal alkanes is thermodynamically favored under solar nebula conditions only if the formation of graphite and CH_4 is kinetically inhibited. Zolotov and Shock (2001) revisited the thermodynamic meta-stability of high-molecular-weight organic molecules and found that hydrocarbons can form in a narrow region of the solar nebula that moves inward as the accretion rate of gas through the disk decreases with time. Low temperatures, high CO abundance, low H_2O abundance, and low viscous efficiencies favor stability of hydrocarbons. Kress and Tielens (2001), using a kinetics model, likewise concluded that Fischer–Tropsch reactions would have occurred only within a narrow region of the solar nebula in the vicinity of the asteroid belt. Sensitivity to nebular pressure and temperature conditions would suggest large variations in planetary C and N abundance between systems. Gaidos (2000) showed that the ratio of C to O varies from 0.3 to >1 in solar-type stars (the solar value is about 0.5). A nebula with higher C/O would be CO-rich and H_2O-poor, possess a more reducing chemistry amenable to organic synthesis, and produce more C- and N-rich planets. However, a recent measurement of a very light ($\delta^{13}C < -100‰$) isotopic composition of solar wind C embedded in lunar regolith grains implies that the formation of chondritic organics included a fractionating step that enriches for ^{13}C (Hashizume *et al.,* 2004). Fischer–Tropsch reactions enrich for ^{12}C. The requirement for ^{13}C enrichment, along with the observed D enrichment, suggests that this process was not a significant contributor to planetary C in our Solar System (Hashizume *et al.,* 2004). Robert (2002) described an alternative scenario in which organics were produced in ion-molecule reactions driven by x-ray irradiation of the outermost layers of the protosolar nebula. These reactions are capable of producting the observed D enrichment.

Similar arguments apply to the origin of planetary nitrogen. Owen *et al.* (2001) used mass spectrometer data returned by the Galileo entry probe to estimate $\delta^{15}N = -370 \pm 80‰$ in Jupiter's atmosphere. Similar values are found for solar wind N implanted in lunar soil (Hashizume *et al.,* 2000, 2004), which implies enrichment of planetary and meteorite organics in ^{15}N. Charnley and Rodgers (2002) explained this enrichment by arguing that conditions in molecular cloud cores are conducive to the formation and condensation of ammonia (NH_3). They calculated that NH_3 ice will be isotopically enriched by almost 80% relative to the (cosmic) value in N_2 (equivalent to an increase in $\delta^{15}N$ of $+200‰$). They proposed that NH_3 ice, rather than N_2, was the principal source of planetary N and speculated that NH_3 was processed into more refractory organic species by ultraviolet or particle radiation. Aléon and Robert (2004) used the observed fractionation of H and N in meteorite and interplanetary dust particle organics to estimate the temperature dependence of the isotopic exchange reactions and equilibration temperatures. The temperature dependence is consistent with the calculations of Terzieva and Herbst (2000), while the temperature range (50–80 K) agrees with those of "hot" molecular cloud cores. This suggests that conditions in progenitor cloud cores, rather than disks, might determine the abundance of N in planet-forming material.

Planetary inventories of C and N are also determined by the efficiency of incorporation during accretion and their partition amongst the surface, mantle, and core during differentiation and subsequent evolution of the planet. Once the proto-Earth had grown large enough, impact-degassed volatiles would have accumulated in a primordial atmosphere containing H_2, CO_2, N_2, and H_2O (Matsui and Abe, 1986). The efficiency of impact degassing increases with planetary mass, as does the ability of a planet to retain gases against lost to space. Tingle (1998), using petrologic and experimental studies of shocked CC meteorites, estimated that the efficiency of C incorporation during accretion on Earth was 1–10%, consistent with the whole Earth $C/^{36}Ar$ ratio compared with CC meteorites.

A hydrodynamic flow of hydrogen escaping from an upper atmosphere heated by the extreme

ultraviolet radiation of the parent star or giant impacts can carry away C and N atoms as well. Direct impact "erosion" (Melosh and Vickery, 1989) occurs only on planets much less massive than the Earth (Ahrens, 1993). For example, simulations by Newman *et al.* (1999) showed that an impact the size of the Cretaceous–Tertiary boundary event results in a net gain of volatiles to the planet. However, this may not hold for the largest (Moon-forming) events (Chen and Ahrens, 1997; Genda and Abe, 2003). In contrast, hydrodynamic escape can explain the non-solar (and non-meteoritic) distribution of the non-radiogenic isotopes of terrestrial rare gases (Porcelli and Pepin, 2000). [These patterns can also be explained, at least in part, by a cometary contribution to the volatile inventory of Earth (Owen and Bar-Nun, 2000).] However, terrestrial N does not have the expected large ^{15}N enrichment. One possibility is that the primordial atmosphere was completely lost by hydrodynamic escape and that the majority of terrestrial nitrogen was degassed from the mantle after the epoch of large impacts and high solar extreme ultraviolet radiation.

Carbon and nitrogen can also be sequestered in the interior of a terrestrial planet. Surface volatiles could have dissolved into the magma ocean created by a giant impact or core formation (Podesek and Ozima, 2000). Carbon, a siderophilic element, can be scavenged by Fe melt and sequestered into the core during giant impacts and core formation (Abe, 1993; Wood, 1993). Kuramoto and Matusi (1996) estimated the partitioning of C between silicate and metallic Fe melts and the fluid phase in the H-C-O system at upper mantle pressures and temperatures. They analyzed a "standard case" (2,000 K and H/C = 5), and three additional cases with different temperatures or H/C ratios. In all four cases they predicted that 70–100% of C is alloyed into molten metallic Fe; the remaining C is in the fluid as CO, CO_2, or CH_4. On this basis they argued that a significant fraction of Earth's initial C was sequestered in the core and that this is responsible for the high H/C and low C/^{36}Ar of the mantle with respect to meteorites. If planetary cores are a significant C sink, then factors that influence core size such as the primordial Fe/Si ratio or the amount of Fe in reduced or metallic form may also control the amount of residual C at the surface. Kuramoto (1997) attempted to explain differences between present Earth and Mars in terms of the relative amount of C sequestered in their cores. The sulfur-rich core of Mars could capture less C, leading to a higher mantle abundance of C, reduction of H_2O to H_2 (which escapes to space), and a higher martian CO_2/H_2O ratio.

Recent experimental and theoretical work bears on the sequestration of C and N in the interior of the Earth. The solubility of N_2 in silicate melts is of particular interest because the mantle ratio of N_2 to ^{36}Ar (both ostensibly inert) is much higher than the atmospheric value and because early experiments showed high solubility under strongly reducing conditions. Two groups have found that solubility increases markedly at an oxygen fugacity (oxidation state) below the iron–wüstite buffer, possibly because N dissolves as N^{3-} rather than N_2 (Libourel *et al.,* 2003; Miyazaki *et al.,* 2004). Earth's mantle was probably more oxidizing than iron–wüstite over most of its history, except before segregation of metallic Fe into the core. These findings raise the possibility that planetary mantle chemistry influences total N inventory. N_2 is sequestered from a primordial planetary atmosphere, but later degassed into a secondary atmosphere from a more oxidized mantle.

A possible reservoir of C in the Earth's core can be constrained by the discrepancy between the experimentally determined density of pure iron at core conditions and the seismically measured density of the outer core (Anderson and Isaak, 2002). This prospect has motivated high-pressure experiments on binary and ternary systems involving Fe and C (Hilgren *et al.,* 2000). Scott *et al.* (2001) found that formation of cementite (also called cohenite) (Fe_3C) from Fe and magnesite ($MgCO_3$) is thermodynamically favored under mantle conditions. They have also experimentally determined that Fe_3C is stable at outer core pressures and calculated that it could produce the core's density anomaly. But is core C imprisoned in perpetuity? Tingle (1998), citing the similar isotopic composition ($\delta^{13}C = -5‰$) of juvenile terrestrial mantle C and graphite in iron meteorites, suggested that the source of the former is C that diffused across the core–mantle boundary and was transported to the upper mantle in plumes.

Volatiles not sequestered in the core or lost to space are exchanged between the mantle and surface via degassing at zones of mantle melting (mid-ocean ridges) and zones of crustal recycling (subduction zones). The tempo of this geochemical cycling is critical to habitability. For example, the abundance of mantle C will affect the flux of

volcanic CO_2, which, if balanced by silicate weathering-limited carbonate formation, will control the surface temperature of the planet. The C content of mantle source regions can be inferred from estimates of the initial CO_2 content of mid-ocean ridge magmas. Holloway (1998) derived a lower limit of 40–80 ppm by weight in this way. Saal *et al.* (2002) estimated a C (as CO_2) content of 72 ± 19 ppm in the upper mantle using basalts that are undersaturated in volatiles (and hence un-degassed). These estimates are equivalent to a mantle C reservoir 0.4 to four times the surface reservoir (Coltice *et al.,* 2004). Solubility of C in olivine (the principal mantle mineral) is <1 ppm by weight at upper mantle pressures (Keppler *et al.,* 2003), and thus mantle C must exist as a separate phase such as carbonates. Isshiki *et al.* (2003) reported that $MgCO_3$ and a previously unknown, high-pressure form of the mineral are stable in the lower mantle. Fiquet *et al.* (2002) calculated that $MgCO_3$ does not dissociate anywhere in the modern terrestrial mantle except perhaps in the core–mantle boundary. Magnesite may represent a significant reservoir of C in the modern Earth. In younger (hotter) or more massive planets the mantle adiabatic temperature profile might cross the stability boundary of $MgCO_3$. If there is a significant fraction of Earth's carbon in the lower mantle as $MgCO_3$, then the absence of this reservoir in other planets could mean a higher abundance in the upper mantle and crust, higher concentrations of dissolved CO_2 in magmas, and a higher volcanic CO_2 flux.

The modern flux of "juvenile" carbon (as opposed to recycled carbon) from the mantle can be estimated either by scaling to the flux of helium (He) or by combining estimates of the global production of mantle melt and the mantle carbon content. Sano and Williams (1996) used measurements of $^{13}C/^{12}C$, $CO_2/^3He$, and $^3He/^4He$ at subduction zones to estimate the total flux of mantle carbon from mid-ocean ridges, hot spots, and subduction zone (where the recycled carbon has been subtracted) to be 1.8×10^{12} mol year^{-1}. Holloway (1998) estimated the mid-ocean ridges flux alone to be 3–6 $\times 10^{12}$ mol year^{-1}, sufficient to supply surface inventory of about 60 bars of CO_2 in 3.3 billion years. A corollary of the latter estimate is that the early Earth had much less carbon on its surface. However, Sleep and Zahnle (2001) used an updated global arc volume from Plank and Langmuir (1998) to derive a juvenile carbon flux two to three times lower than that of Sano and Williams (1996), in which case most of the surface carbon reservoir would be "primordial," perhaps from impact degassing

Stable isotopes again offer insight into the long-term evolution of Earth's carbon cycle. Coltice *et al.* (2004) attempted to reconcile the fact that isotopically heavy ($\delta^{13}C \approx 0‰$) carbonates are preferentially subducted with respect to light ($\delta^{13}C \approx -25‰$) organics with the observation that mantle $\delta^{13}C$ has not deviated much from over −5‰ geologic time. That reconciliation requires that the subducted C is sequestered, *e.g.,* as slabs at the base of the mantle, or that there is a deep C-rich reservoir that feeds the mantle, *e.g.,* the core. However, the latter hypothesis requires "fine tuning" to maintain a constant mantle $\delta^{13}C$. Thus, the fate of C (primarily as carbonates) in subducted marine sediments is an important link in the Earth's C cycle. Devolatilization of crust at subduction zones could return C as volcanic CO_2 to the ocean-atmosphere; carbonate phases stable at the pressures and temperatures experienced by descending slabs of oceanic lithosphere would represent a sink of surface C. Hammouda (2003) reported the results of high-pressure melting experiments with glassy basalt–carbonate–water mixtures and found that transition to an eclogite + carbonatite melt phase and removal of CO_2 are inevitable below 300 km, except perhaps in carbonate veins in deep fractures of the crust. These results raise the possibility that carbon in the lower mantle is entirely primordial.

The terrestrial nitrogen cycle is more poorly described. The surface inventory (3.5×10^{20} mol of N, equivalent to 1.0 bar of N_2) consists mostly of atmospheric N_2 with the balance in sedimentary and crystalline rocks (Fig. 5). Degassing of the mantle at mid-ocean ridges and subduction zones introduces N to the Earth's surface, mostly as N_2. Marty (1995) measured the ratio of N_2 to ^{40}Ar (gases with similar behavior) in mid-ocean ridge basalts and used the known mantle flux of He and ratio $^4He/^{40}Ar$ to calculate the flux of N and the concentration of mantle N. He estimated that the amount of N in the mantle is roughly equal to that on the surface and that the flux of volcanic nitrogen from mid-ocean ridges is $2.2 \pm 1.0 \times 10^9$ mol of N_2 year^{-1}. Sano *et al.* (2001) combined Marty's estimate with their own for island arcs, back-arc basins, and intraplate hot spots to estimate a global volcanic N flux of 2.8×10^9 mol of N_2 year^{-1}. Integration of this flux over the age of the Earth is equivalent only to ~7% of the cur-

rent surface inventory, which suggests higher rates of early N degassing.

Unlike CO_2, N_2 is chemically inert and removed from the atmosphere only by biological fixation into organic form and, to a much lesser extent, conversion into soluble NO_x molecules by lightning. Nitrogen is incorporated as organic matter in seafloor sediments and returned to the mantle or degassed back to the atmosphere at subduction zones. The high N_2 content of the venusian atmosphere tempts a speculation that, in the absence of single nitrogen species produced by life or reaction with oxygen, the return of nitrogen to the mantle would occur at a much lower rate, and almost all of the nitrogen inventory of the Earth (~2 bars) would reside in the atmosphere. However, the estimates of Sano *et al.* (1998, 2001) of the contributions of mantle and sediment sources to the flux at mid-ocean ridges, island arcs, back-arc basins, and hot spots indicate that only one-third of the total flux is from recycled sediments. Thus the recycling of N by plate tectonics has not contributed significantly to the surface inventory, and N remains in the mantle because of incomplete degassing, rather than efficient recycling. Venus's N_2-rich atmosphere may be the signature of recurrent magma oceans and efficient degassing. It would thus appear that events early in an Earth-like planet's history exert the dominant influence on the N_2 content of its atmosphere.

LESSONS FROM A LOST WORLD

No review of planetary habitability would be complete without a discussion of our only calibration point: the appearance of habitable conditions and life early in Earth's history. That early history was marked by dramatic events, which include core formation, the Moon-forming impact, and a possible magma ocean. Accretion, degassing, and massive hydrogen escape were probably complete by 4.47 billions of years ago (Ga), and left behind an anoxic atmosphere of N_2 and CO_2 (Podesek and Ozima, 2000). The first primitive crust formed by 4.4 Ga (Amelin *et al.*, 1999), and the atmosphere was closed to all elements except H and He by 4.3 Ga (Tolstikhin and O'Nions, 1994) (Fig. 6). We describe recent work that constrains the appearance of habitable conditions and life, or challenges previous conclusions. Readers are referred to Kasting and Catling (2003) for a comprehensive review of Earth's biogeochemical evolution.

The formation of a hydrosphere, a crucial step in habitability, is a consequence of evolving physiochemical conditions on or near the surface of the planet (we discuss the source of terrestrial planet water later). The timing, physical state, and chemistry (pH, alkalinity, and calcium ion concentration) of the hydrosphere will constrain the tempo and nature of early life. Sedimentary rocks bear witness to Earth's past hydrological cycle and standing bodies of water. The oldest known rock—4.03 Ga Acasta gneisses—are heavily metamorphosed, and the nature of the parent rocks cannot be discerned (Bowring and Williams, 1999). The evidence for more ancient rocks consists of detrital zircons in metamorphosed sediments (Amelin *et al.*, 1999). Zircons from the Jack Hills in the Narryer Gneiss Terrane (Western Australia) have an average $^{207}Pb/^{206}Pb$ age of 4.3 Ga (Mojzsis *et al.*, 2001; Wilde *et al.*, 2001); the oldest of these is 4.404 ± 0.008 Ga (Fig. 6). Some of the zircons derive from parent magmas that are enriched with the heavy isotope of oxygen relative to the mantle with $\delta^{18}O$ values of 7.4–9.7‰ compared with 5.5‰ ($\delta^{18}O = [(^{18}O/^{16}O_{sample})/(^{18}O/^{16}O_{SMOW}) - 1] \times 1{,}000$. This enrichment is indicative of melting and assimilation of high-$\delta^{18}O$ rock that interacted with water at low temperatures, *i.e.*, on or near the surface.

These findings suggest the early existence of a hydrosphere; however, the zircons do not provide information on the extent or physical state of that hydrosphere, which, rather than open oceans, could have been confined to the subsurface or covered by ice. Even if oceans existed they may have been inhospitably hot (~200°C) under a thick CO_2/H_2O atmosphere. The oldest unequivocal sedimentary rocks are from the 3.85 Ga Isua banded iron formation in Greenland (Nutman *et al.*, 1997). At earlier times, higher levels of atmospheric CO_2 or methane (CH_4) may have compensated for a fainter young Sun and maintained surface temperatures above the freezing point. However, Sleep and Zahnle (2001) pointed out that CO_2 would have been removed from the atmosphere by weathering of seafloor and impact ejecta produced by intense post-accretion bombardment. In the absence of another greenhouse gas such as CH_4, Earth's surface would have been cold, and any ocean ice covered.

The chemistry of an abiotic primitive hydrosphere would have been controlled by atmo-

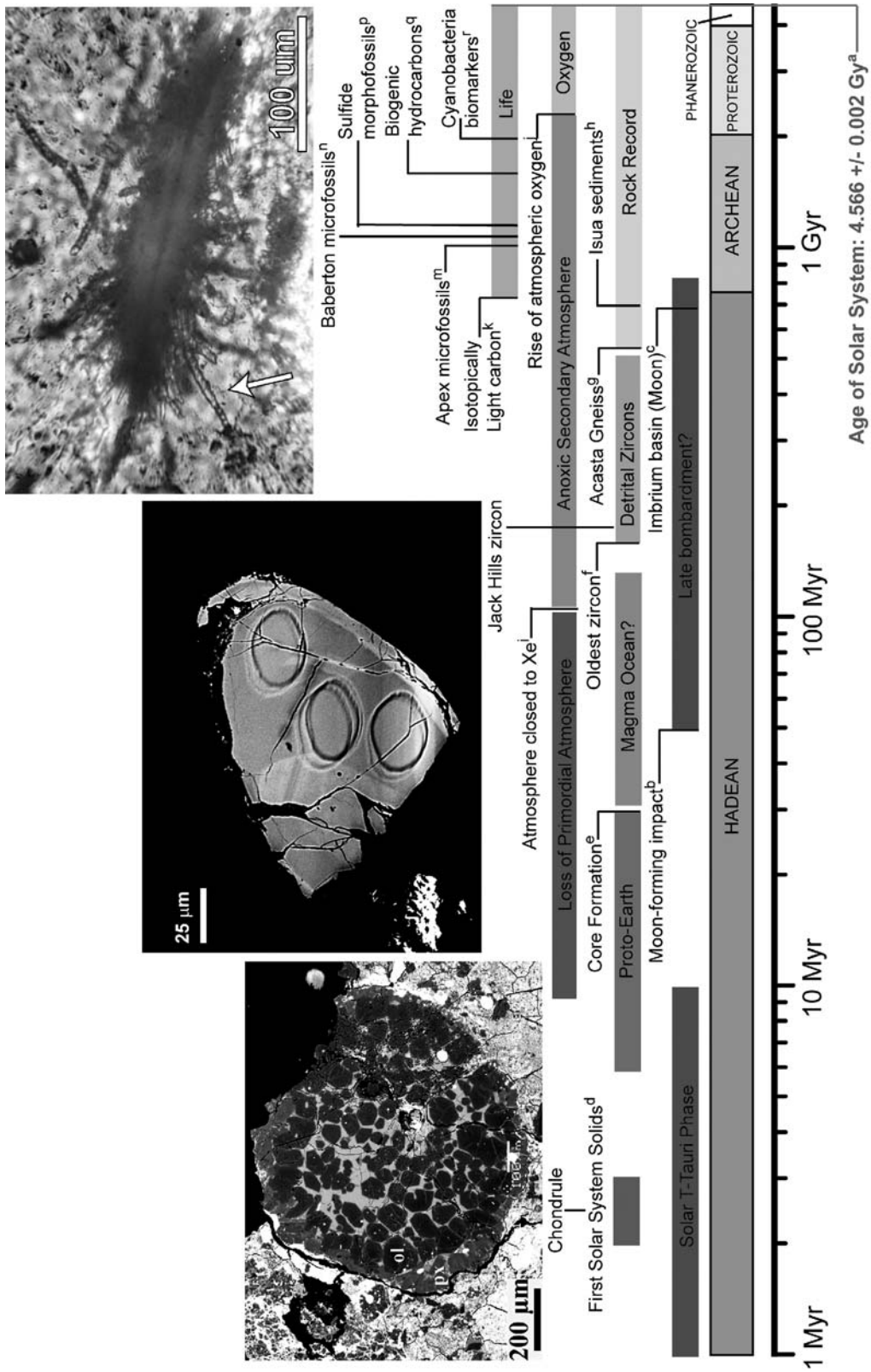

FIG. 6. History of the Earth on a logarithmic time scale to emphasize the number of important events at early times and our relative ignorance of the timing of those events. Times are measured with respect to closure for radiogenic Al-26 or U in the early Solar System. The gray bar denotes geologic ages, and the colored bars above represent Solar System events and geologic, atmosphere, and biological evolution on the Earth. Images are, from left to right: backscattered electron image of a magnesian (Type I) chondrule from a CR carbonaceous chondrite (ol = olivine, py = pyroxene) (courtesy of Alexander Krot); backscattered electron image of zircon JH992-76 from the Jack Hills with a near concordant Pb^{207}/Pb^{206} age of 4.042 ± 0.015 Ga (courtesy of Stephen Mojzsis); and micrograph of features of microbial origin in pillow lavas from the 3.45 Ga Barberton Greenstone Belt (Furnes *et al.*, 2004) (courtesy of Neil Banerjee and reproduced with permission from *Science*). References are as follows: a, Allègre *et al.* (1995); b, Lee *et al.* (2002); c, Deutsch and Stoffler (1987); d, Amelin *et al.* (2002); e, Kleine *et al.* (2002); f, Wilde *et al.* (2001); g, Bowring and Williams (1999); h, Nutman *et al.* (1997); i, Podosek and Ozima (2000); j, Rye and Holland (1998); k, Mojzsis *et al.* (1996); m, Schopf (1993); n, Furnes *et al.* (2004); p, Rasmussen (2000); q, Dutkiewicz *et al.* (1998); r, Brocks *et al.* (1999).

spheric CO_2 concentration, weathering reactions, and any input of hydrothermal fluids. Two scenarios are proposed for early ocean chemistry: an acidic to neutral chemistry (Garrels and Mackenzie, 1971) and an alkaline chemistry (Kempe and Degens, 1985). Garrels and Mackenzie (1971) proposed that volatiles were rapidly degassed during accretion, which produced a secondary atmosphere rich in H_2O, CO_2, and HCl. As the Earth cooled, HCl would have dissolved in seawater and reacted with CO_2 and the crust. This would have produced a chloride ocean in which chlorine (Cl^-) ions were balanced by sodium (Na^+) and calcium (Ca^{2+}) ions in roughly equal quantities, with lesser concentrations of potassium (K^+) and magnesium (Mg^{2+}) ions. Carbonation of the solution would have removed Ca^{2+} as calcite or aragonite ($CaCO_3$), and weathering of Na^+ from the crust would maintain charge balance with Cl^-. The alkaline ocean model was developed by analogy with modern soda lakes in active volcanic regions. Kempe and Degens (1985) assumed that while H_2O, CO_2, and CH_4 were degassed at an early epoch, HCl was released more slowly. In their scenario, rapid weathering of silicate rocks prevented the formation of a CO_2-rich atmosphere and created a high pH ocean dominated by Na^+, carbonate (CO_3^{2-}), and bicarbonate (HCO_3^-), with a lesser concentration of Mg^{2+}. Calcium would have been depleted by carbonate precipitation. However, Sleep *et al.* (2001) argued for a scenario where a CO_2-rich atmosphere created by the Moon-forming impact is stabilized by the slow rate at which CO_2 can be returned to the early mantle (if plate tectonics is not operating). They also pointed out that the low Na:Al of the Earth means that soluble Na minerals are stable only at atmospheric CO_2 levels higher than the total planetary inventory. Furthermore, atmospheric CO_2 must have built up to levels sufficient to maintain liquid water at the surface.

None of these scenarios is constrained during the Hadean (prior to 3.8 Ga), but there is information about the Archean (3.8–2.5 Ga) atmosphere from the rock record. Rye *et al.* (1995) inferred an upper limit of 0.04 bars for atmospheric CO_2 based on the absence of iron carbonates in 2.75–2.2 Ga paleosols. On the other hand, Hessler *et al.* (2004) used the presence of iron-rich carbonate relative to clay minerals in 3.2 Ga river gravels to set a minimum partial pressure of CO_2 of 0.0025 bars at 25°C (higher at warmer temperatures). Grotzinger and Kasting (1993) estimated an Archean ocean pH of between 6 and 8.5. They interpreted the lack of gypsum ($CaSO_4$) in evaporites before 1.7 Ga as indicating low ocean calcium concentrations; calcium ions were removed by precipitation as carbonate. They assumed that the ocean was saturated with respect to siderite ($FeCO_3$) as well as $CaCO_3$ (justified by the appearance of siderite in limestone facies) and that soluble ferrous iron (Fe^{2+}) was 1.0 ppm (justified by the existence of banded iron formations). Archean rocks contain massive carbonate formations that would not have formed (abiotically) unless the oceans were supersaturated with respect to calciate by a factor of 20 (Morse and He, 1993). Grotzinger and Kasting (1993) calculated [Ca^{2+}] based on the absence of gypsum, used calcite supersaturation to find [CO_3^{2-}], and assumed 0.3 atm or 0.03 atm partial pressure of CO_2 to find the pH. Their calculations are consistent with an upper limit on the total amount of early surface carbon (Walker and Brimblecombe, 1985). A higher pH is permitted if CO_2 levels were much lower or siderites were metasomatized iron oxides. Morse and Mackenzie (1998) used a Pitzer model for the carbonic acid system in seawater modified by He and Morse (1993) to estimate a pH of 6.8 for the late Hadean ocean. They assumed a seawater temperature of 85°C, 1 bar of atmospheric CO_2, and a calcite saturation state of 20.

The salt concentration of the early ocean is also debated. Morse and Mackenzie (1998) and Hardie (2003) argued that the ocean was always an NaCl-dominated aqueous solution. They assumed that all salts in modern evaporate deposits were present in the Hadean ocean in dissolved form, which resulted in a salinity twice that of today. In contrast, Kempe and Degens (1985) suggested that the chloride concentration was initially much less than the bicarbonate concentration and that chloride steadily accumulated in the ocean as bicarbonate precipitated as carbonates. The chloride content of the early ocean would have been influenced by the ratio of HCl/carbon in released volatiles and the degassing history of the Earth. The HCl/carbon ratio estimated by Garrels and Mackenzie (1971) is in agreement with today's ratio, whereas Kempe and Degens (1985) estimated that the gases degassed were initially C-rich and would have become HCl-rich over time. In this context, the extensive degassing associated with the Moon-forming impact may have favored the formation of a saline ocean.

Microfossils, microbially influenced sedimentary deposits (*e.g.,* biogenic stomatolites), and chemofossils (carbon-isotopic composition and biomarker presence) constitute the evidence for the earliest life on Earth. Doolittle *et al.* (1996) proposed that the major groups of living things arose by 3.5 Ga based on the sequence divergence of proteins in extant lineages, but this chronometry is sensitive to calibration errors and the assumption of a molecular "clock." The abundance and quality of biosignatures have been degraded by the intense metamorphism of the host rocks, which complicates interpretation. The oldest purported microfossils consist of cyanobacterium-like microorganisms in the 3.45 Ga Apex cherts (Schopf, 1993). However, Brasier *et al.* (2002) suggested an alternative formation mechanism for the microfossil-like structures. They described evidence of hydrothermal origin and alteration of the host cherts rather than a sedimentary origin. Brasier *et al.* (2002) argued that the carbon isotopic fractionation of the carbonaceous material within the Apex chert, while not inconsistent with biogenic fractionation, is also found in other hydrothermal cherts. They reported that the microfossil-like structures occur in secondary rocks and are associated with volcanic shards, calling into question their biogenicity. The two reports disagree regarding whether the two peaks that appear in Raman spectra of the features represent two forms of amorphous carbon (graphite) or kerogens. Altermann and Kazmierczak (2003) favored a biogenic origin, pointing out that any abiotic mechanism must be able to produce the organic carbon compounds, the isotopically light carbon, and the microfossil-like structures.

In a recent publication, Furnes *et al.* (2004) have described micron-sized tubes in 3.5–3.2 Ga pillow lava of the Barberton Greenstone Belt (Fig. 6). They interpreted the morphology of these structures, combined with the presence of carbon, nitrogen, and nucleic acids within the altered igneous glass, as indicating a biogenic (microbial etching) origin. The next oldest microfossil evidence consists of pyritic filaments in a 3.2 Ga deep-sea volcanogenic deposit analogous to modern deep-sea hydrothermal systems (Rasmussen, 2000).

Macroscopic biogenic structures such a stromatolites formed by the activity of microbial mat communities are also indicators of the past life. Although microbes are undoubtedly involved in modern stromatolite formation, Archean examples rarely contain microfossils. Grotzinger and Rothman (1996) questioned the assumption that biological processes are required for all past stromatolite construction. They described four mechanisms of abiotic stromatolite formation and preservation: sediment fallout, smoothing of settled sediment, surface-normal precipitation, and random noise. They did not reject the possibility of life existing at 3.5 Ga, but caution against the use of stromatolites as definitive and sole indicators of life. Challenges to the biogenicity of 3.5 Ga microfossils, therefore, cast additional doubt on the nature of 3.5 Ga stromatolites.

Chemofossils such as carbon isotopes and uniquely biogenic material are less ambiguous and can be better preserved than morphological fossils. Mojzsis *et al.* (1996) measured isotopically light carbon ($\delta^{13}C = -30‰$, $\delta^{13}C = [(^{13}C/^{12}C_{sample})/(^{13}C/^{12}C_{standard}) - 1] \times 1{,}000$) values for graphitic inclusions within apatite grains in the 3.8 Ga Greenland Isua banded iron formation. They concluded that these values indicate biological activity consistent with photoautotrophic bacterial metabolism, rejecting the hypothesis that organic matter was thermally metamorphed, which would have released isotopically light CH_4 and heavy CO_2. Van Zuilen *et al.* (2002) constructed an alternate view of graphite genesis and cast doubt on the conclusions drawn by Mojzsis *et al.* (1996). Van Zuilen *et al.* (2002) challenged the sedimentary origin of the host rocks and pointed out that coexistence of graphite with iron carbonate in the Isua sequence is evidence for metasomatism, an abiotic process that forms secondary material by reaction of hot fluids with older crustal rocks. However, not all the data obtained by van Zuilen *et al.* (2002) (such as the $\delta^{13}C$ values) exclude a biogenic origin for the graphite. Furthermore, they did not analyze graphite contained in apatite crystals, which can act as a barrier for isotopic and material exchange with surrounding fluid and rock.

The next oldest chemofossils evidence is oil preserved as 5–10-μm fluid inclusions within detrital quartz grains of 3 Ga Archean sandstones (Dutkiewicz *et al.,* 1998). These hydrocarbons are derived from the thermal maturation of biogenic kerogen and establish the presence of aquatic biota by 3 Ga. The rarity of Archean oil inclusions could reflect low productivity of the early biosphere, high productivity and concurrent high aerobic remineralization, or low preservation rates. Specific biomarker molecules produced by

diagenesis of unique progenitor biomolecules can be used to establish the diversity of life on early Earth (Dutkiewicz *et al.,* 1998; Simoneit, 2004). Detection of 2α-methylhopanes, hydrocarbons uniquely linked to extant cyanobacteria, suggests the existence of oxygenic phototrophs by 2.7 Ga (Brocks *et al.,* 2003).

The timing and nature of early life on Earth are important, if nominal, constraints on models of early terrestrial planetary habitability. The difference between life appearing after 0.3 Ga—corresponding to the earliest evidence for habitable conditions on Earth—and after 1.8 Ga—corresponding to a non-contentious minimum age for the appearance of life—has direct implications for selection of stars that are considered sufficiently long-lived to host habitable planets. Also, a biota based on oxygenic photosynthesis may develop only under specific ranges of temperature and atmospheric CO_2 (Franck *et al.,* 2001).

DE FONTANELLE'S DILEMMA

We have considered some of the factors responsible for the habitability of planet Earth, including a semimajor axis within the HZ, geologic activity, abundant water, and sufficient inventories of carbon and nitrogen. The recent work reviewed here suggests that: (1) Earth-size planets will commonly form in the HZ of solar-type stars, (2) many of these planets will experience active geologic re-surfacing at some point during their history, (3) a large fraction of them will have an initial water inventory equal to or larger than the Earth, and (4) the abundance of carbon and nitrogen may vary markedly among these planets. What is not understood is how synergisms between these factors affect habitability. Simulating the dynamics and evolution of a habitable planet is a daunting task (Meadows *et al.,* 2001), and significant progress may have to wait until we have more than one example of a habitable planet. The "difficulty of figuring it out" will drive future science in this new century, and will hopefully be the basis for a series of ambitious missions to detect and characterize extrasolar terrestrial planets. The Kepler Mission, scheduled for launch in 2006, will be designed to detect the transits of Earth-sized planets around any of ~100,000 stars (Borucki *et al.,* 2003). Several architectures for space missions to detect extrasolar terrestrial planets in either reflected (optical) or emitted (infrared) light are under consideration (Beichman *et al.,* 2002; Brown *et al.,* 2003; Fridlund and Gondoin, 2003; Levine *et al.,* 2003; Lille *et al.,* 2003). These missions are designed to obtain photometric and spectral information that can be used to infer the presence of thick atmospheres, clouds, or water on these planets, and even biological pigments on their surfaces. A more comprehensive picture of planetary habitability will undoubtedly emerge from these observations, along with an advanced understanding of the origin and evolution of our own world and the nature of its ability to support a biota.

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ABBREVIATIONS

AU, astronomical units; CC, carbonaceous chondrite; Ga, billions of years ago; HZ, habitable zone; MEGNO, Mean Exponential Growth factor of Nearby Orbits; Myr, million years; SMOW, standard mean ocean water.

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